Response to the review from the Journal of Forecasting

Manuscript 7441141, "Online Anchor-Robust Forecasting under Economic Regime Shift"

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I thank the reviewer for a careful and constructive report. The manuscript was declined at the *Journal of Forecasting* and has been revised before this transfer. Below, the reviewer's comments are in italics, each followed by the change made. Section, table and page numbers refer to the revised manuscript.

Main concern: the paper is difficult to follow

> *My main concern, however, is that the paper is unusually difficult to follow. Despite my familiarity with regime-switching models, I struggled to connect the economic motivation, the assumptions about the anchor, and the empirical exercise. The paper introduces several extensions before establishing why the basic method is needed.*

I agree, and I have reorganised the whole paper around this point.

- The introduction now states one question: if a regime shift moves an observed driver, can the forecaster be made robust to it in advance, and what does that cost when no shift occurs?
- The basic method, with the anchor supplied by the analyst, and its guarantee are presented first (Section 3), with no mention of learning the anchor.
- The simulation study (Section 4) and the energy application (Section 5) both use the basic method.
- Learning the anchor from data and the eigengap diagnostic were previously mixed into the method section. They are now a separate extension (Section 6), placed after the main results and introduced as the answer to a practical question raised by Section 5: what to do when the analyst does not know which driver carries the shift.
- Section 5 has a new paragraph, "The anchor and what it assumes". It states which indices are used as the anchor, what would have to be true of them for the method to help, why the news-based indices are more plausible candidates than the VIX, and what the stress tests of Section 4 predict if the anchor is only weakly aligned with the true channel. This connects the assumptions of Proposition 1 to the empirical exercise.
- The conclusion (Section 7) is shorter. It ends with a practical procedure for deciding whether to use the method.

> *A simple forecasting example near the beginning would help explain what changes across regimes, what stays stable, and why the proposed method should improve forecasts.*

The introduction now contains such an example (paragraph "An example", Table 1).

- A target depends on a stable predictor. A second predictor tracks the target but is also moved by an observed driver, in the way a news-based indicator might rise with volatility and also with the volume of geopolitical headlines.

- In the estimation sample the driver has two regimes; in the forecast period it moves to a level not seen before.
- The example states explicitly what changes (the relation between the target and the second predictor) and what stays stable (the relation with the first predictor).
- Least squares puts weight on the second predictor, has the lower in-sample error, and roughly doubles its error in the new regime (0.206 to 0.452).
- Anchor regression removes that weight and loses almost nothing in the new regime (0.240 to 0.242), at a small in-sample cost.
- The example also shows what the method needs: if the analyst chose a driver unrelated to the shift, the penalty would remove signal and buy no protection.
- The script that produces Table 1 is in the replication files (`examples/two_signal_example.py`).

> *The paper needs a much clearer account of its purpose and contribution before its results can be properly assessed.*

The paragraph "What the paper does" in the introduction now lists the contribution section by section. The paragraph that follows states the findings, including the negative ones:

- the gain in simulation disappears when the anchor is noisy or misaligned;
- on the energy data OARF is not better than online gradient descent or an equal-weight average;
- the learned anchor fails on a minority of streams.

The abstract was rewritten in the same way.

Technical note: the overlapping target

> *The target on pg. 10 is the log of a trailing five-day average of squared daily returns. At the stated one-step forecast horizon, four of the five squared returns are already observed. This creates substantial mechanical persistence, which needs to be considered when interpreting the forecast results.*

The reviewer is right, and I confirmed it in the code. The old target shared four of its five terms with its own lag, and on Brent its correlation with the previous day's value was 0.89. The real-data study has been rebuilt and rerun with a target that does not overlap (Section 5, "Data and target").

- *Target.* The target is now the log of weekly realised variance, $y_w = \log \sum_{s \in w} r_s^2$, over the daily returns of the week ending on Friday w.
- *Information set.* Every forecast of y_w uses data up to the end of week $w-1$ only.
- *No overlap.* Consecutive weeks share no terms. The correlation of the target with its own lag falls to 0.53.
- *Sample.* There are 847 weeks, of which 678 are evaluated.

- *Forecast pool.* The pool was rebuilt for the weekly frequency: last week's value, HAR-type 4- and 13-week means, RiskMetrics and GARCH variances, the lagged VIX, and a rolling ridge regression.
- *Pre-processing.* The drivers are no longer pre-scaled with full-sample statistics, so all scaling now uses past data only.

The earlier daily design is kept in Appendix B, with Table 11, so the reader can see how much the overlap flattered every method.

> *A simple persistence forecast should be reported alongside the proposed method so that the reader can see what is gained beyond the target's existing predictability.*

Persistence (last week's value) is now reported in Table 9, together with the equal-weight average of the forecast pool, and both are scored under the same protocol as the learning methods. The two designs give different answers.

- *Weekly design.* Every learned combination beats persistence. For most methods the MSE is about a third lower on Brent and a fifth lower on Henry Hub. The Model Confidence Set excludes persistence on Brent ($p < 0.01$). On Henry Hub persistence stays in the set ($p = 0.17$), and the text says that the gain there is not significant.
- *Daily design.* In the old daily design persistence is inside the Model Confidence Set with $p = 1.00$ on both series (Appendix B). This confirms the reviewer's concern that the earlier results mostly reflected the mechanical persistence of the target.

> *A check using a less overlapping target or a longer horizon would also help establish whether the findings depend on this construction.*

The weekly target has no overlap at all. Under it the conclusion about OARF is the same as before: it is within 1.5% of online gradient descent on both series, and the Diebold–Mariano test of OARF against online gradient descent gives $p = 0.47$ on Brent and $p = 0.54$ on Henry Hub. What changed is the interpretation of the comparison with simple benchmarks, which now shows a real gain from combining forecasts. The paper reads the absence of an OARF advantage as consistent with the uncertainty indices being weak or noisy proxies for the channel, which is the case where the stress tests of Section 4 predict no gain. It also states that the data cannot rule out that energy-volatility regimes are not anchor-mediated mean shifts at all.

Other changes

- The appendices on the volatility-timing evaluation (Appendix C) and on the two event windows (Appendix F) were rerun on the weekly design.
- The text was edited throughout for length and clarity.
- All code and results are public and regenerate every number in the paper.